

Locating and Auditing a CompTIA SY0-701 Study Module in a Personal Obsidian Vault: A Retrieval and Quality-Assurance Case Report

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Abstract

Personal knowledge vaults are frequently used as private study corpora, and the provenance and internal quality of the material stored in them are rarely audited. This paper reports a retrieval-and-audit case study conducted on a single personal Obsidian vault. A case-insensitive recursive search across the vault root `~/Documents/vmain` returned one matching study artifact: *CompTIA Security+ Guide to Network Security Fundamentals* by Mark Ciampa, held in two distinct forms—a 12.2MB source PDF and a directory of extracted per-module notes spanning fifteen modules. The study scope was then narrowed to Module 7, “Identity and Access Management,” aligned to the CompTIA SY0-701 objective domain, and that module was read in full (471 lines, 21.6 KB) and decomposed against its three stated learning objectives: describing authentication credential types, listing authentication best practices, and defining access control schemes. The audit surfaced two concrete defects: an internal contradiction between a stated industry password-expiry practice and an explicit recommendation against expiry, and a malformed review question offering two textually identical answer options. The paper closes by specifying a drill protocol in which a learner supplies examination questions and each is answered with a correctness score and a line-level citation back into the source note. All findings are descriptive of the audited artifact; no empirical learner-outcome data are reported.

1 Introduction

The effective ceiling of a self-directed certification study programme is set by the material the learner can reliably find, read, and trust. For many learners that material does not live in a publisher’s learning platform but in a personal knowledge vault—an Obsidian library that accumulates PDFs, notes, clippings, and self-authored prose without any editorial layer between the source document and the study page. Two practical problems follow. First, a retrieval problem: the learner must locate the authoritative artifact when the same title exists simultaneously as a binary source and as a tree of derived notes. Second, and less often confronted, a trust problem: derived notes are the layer a learner actually reads, yet they are the layer most likely to contain silent inconsistencies, because no editorial pass has verified the derived text against the source or against its own internal consistency.

This paper documents a case study that treats both problems as a single retrieval-and-quality-assurance exercise. The subject is Module 7, “Identity and Access Management,” of a CompTIA Security+ study corpus held in a personal Obsidian vault and aligned to the SY0-701 objective domain.

The contributions of this case report are threefold: (i) a documented account of how a single study title was resolved from a vault-root search into two concrete artifacts, one of which was selected as the study surface; (ii) a descriptive decomposition of the selected module into its objective-aligned content blocks, including the attack taxonomy, the multifactor token distinctions, the biometric modality taxonomy, the digest-protection mechanisms, the single sign-on technology set, and the ranked access control schemes; and (iii) a reproducible quality audit that identified two defects in the note, one logical and one structural. The paper is deliberately descriptive: it characterises an artifact and a process, and it does not claim that the audit improves examination outcomes, because no outcome data were collected.

2 Background and Scope

2.1 The audited corpus

The vault root searched was `~/Documents/vmain`, a personal Obsidian vault on the author's machine. The search was case-insensitive and recursive over the vault root, and it was keyed on the study title rather than on a filename pattern.

Exactly one matching title was returned: *CompTIA Security+ Guide to Network Security Fundamentals* by Mark Ciampa. The title materialises in two forms within the vault:

- (i) a source PDF of 12.2 MB, and
- (ii) a directory of extracted per-module notes covering fifteen modules.

This duality is the substantive reason a retrieval step was needed at all. The PDF is the authoritative container but is poorly suited to targeted revision; the note tree is the revision surface but is a derived artifact whose fidelity is unverified. The case study therefore resolved the ambiguity by taking the note tree as the study surface while retaining the PDF as the provenance reference.

2.2 Module under audit

The learner narrowed scope to Module 7, “Identity and Access Management,” associated with the CompTIA SY0-701 objective domain. The module note was 471 lines and 21.6 KB in size, and was read in full rather than sampled. The module is organised around three stated objectives, quoted here as they frame the study material:

- O1.** describe authentication credential types;
- O2.** list authentication best practices; and
- O3.** define access control schemes.

3 Methods

The procedure had four stages: retrieval, scope selection, close reading, and defect audit. Each stage is reported descriptively.

Stage 1 — Retrieval. A single case-insensitive recursive search was executed over the vault root on the study title. This stage yields the set of candidate artifacts and, by reporting them, exposes the PDF/note-tree duality described in Section 2.

Stage 2 — Scope selection. The learner selected one module from the fifteen available. The selection reduced the audit target from a corpus to a single self-contained note, which made full reading tractable: the entire note was consumed, not sampled, so every claim in Section 4 is grounded in the complete text of a 471-line artifact rather than in a subset of it.

Stage 3 — Close reading and decomposition. The module was read in full and decomposed into content blocks aligned to objectives O1–O3. The decomposition is deliberately shallow and enumerative: it records which concepts the note covers and how they are partitioned, without re-deriving the underlying security content, which is the note’s responsibility and not this audit’s.

Stage 4 — Defect audit. Two classes of defect were checked for. The first is *internal contradiction*: a claim asserted in one part of the note that cannot be reconciled with a claim asserted elsewhere in the same note. The second is *structural malformation*: a comprehension-check item whose answer key is degenerate, for example because two options are textually identical and no unique correct answer can exist. Both checks are properties of the note in isolation and require no external ground truth, which keeps the audit independent of any third-party source.

Stage 5 — Drill protocol. The session concluded with the learner electing a drill protocol rather than a further reading pass. Under this protocol the learner supplies examination questions; the assistant answers each question, assigns a correctness score to its own answer, and cites the supporting lines of the source note. The protocol converts the audited note from a passive reading surface into an active retrieval surface, and the line-level citation requirement makes each answer falsifiable against the artifact rather than merely plausible.

4 Results

4.1 Retrieval outcome

Table 1 summarises the single retrieval stage.

Table 1: Outcome of the case-insensitive recursive vault-root search.		
Search root	Query	Matching titles
~/Documents/vmain	<i>CompTIA Security+</i> (title text)	1

The single matching title resolved to two artifacts, summarised in Table 2.

Table 2: Artifacts materialising the retrieved title, and the role assigned to each.

Form	Size	Extent	Assigned role
Source PDF	12.2 MB	1 file	Provenance reference
Note directory	—	15 modules	Study surface

4.2 Content decomposition of Module 7

Table 3 records the content blocks recovered from the 471-line module, each with the objective it serves. Where the note partitions a concept into a stated number of sub-elements, the count is reported; where individual sub-element names are not recoverable from the audit record, the count is given alone.

Table 3: Content blocks of Module 7 by learning objective. “3+4” denotes a partition of seven elements into three factors and four attributes.

Block	Obj.	Recorded extent
Elements of authenticity	O1	Seven elements, partitioned 3 factors +4 attributes
Password architecture	O1	One-way hashing to message digests
Password attack taxonomy	O1	Eight named classes
Attack sequence	O1	Eight-stage canonical sequence
Multifactor tokens	O1	Time-based vs. event-driven one-time passwords; MFA fatigue
Hardware tokens	O1	Security keys, attestation, PIV smart cards
Biometric modalities	O1	Physiological, cognitive, behavioral
Keystroke dynamics	O1	Behavioral subclass; dwell time and flight time
Digest protection	O2	Salting, peppering, key stretching
Key-stretching functions	O2	bcrypt, PBKDF2, Argon2
Single sign-on	O2	SAML, LDAP, OAuth, OpenID, Shibboleth
Passkeys	O2	Covered as a distinct sign-in mechanism
Access control schemes	O3	Five schemes, ranked by restrictiveness

4.3 Password attack classes

The module names eight password attack classes and additionally specifies an eight-stage canonical attack sequence. Six of the eight classes are recoverable by name from the audit record and are listed in Table 4, separated into the two families the note itself distinguishes: online attacks, conducted against a live authentication endpoint, and offline attacks, conducted against a recovered hash store. The mapping onto families for the rule attack class is not recorded in the audit record and is left unstated rather than inferred.

Table 4: Attack classes named in the module, with family assignment as recorded.

Class	Family as recorded
Brute force (online)	Online
Brute force (offline)	Offline
Password spraying	Online
Dictionary	Offline
Credential stuffing	Online
Rule attack	Not stated in audit record
Two further classes present in the note, names not in audit record	

4.4 Access control schemes

Five access control schemes are covered and are ranked by restrictiveness, from least to most restrictive: discretionary access control (DAC), mandatory access control (MAC), role-based access control (RBAC), rule-based access control, and attribute-based access control (ABAC). The ranking is a single total order; the note does not treat MAC and DAC as equally restrictive, nor RBAC and rule-based as interchangeable.

4.5 Quality audit

The audit surfaced two defects, one of each checked class.

Table 5: Defects identified in the Module 7 note.

Class	Location	Defect
Internal contradiction	Password expiry	The note states an industry password-expiry practice and, elsewhere, explicitly recommends <i>against</i> password expiry. The two positions are not reconciled within the note.
Structural malformation	Review question	A review question presents two answer options that are textually identical, so no unique correct answer is selectable.

The two defects differ in their consequence. The malformed review question is self-announcing: a learner attempting it discovers the defect immediately, and the cost is a wasted item. The password-expiry contradiction is the more dangerous of the two, because it is a genuine internal inconsistency in a normative statement—a practice described as common in industry alongside a recommendation against that practice—and a learner who reads only one of the two passages can absorb either position as the module’s advice. Because the note is a derived artifact, the contradiction may originate in the extraction step rather than in the source PDF; distinguishing those two causes would require re-reading the PDF, which was not part of this case study.

4.6 Drill protocol

The session ended with the adoption of the drill protocol of Section 3: the learner supplies examination questions, and each is answered with an assigned correctness score and a line-level citation into the 471-line source note. The protocol is the intended mechanism by which the defects identified above would surface in practice, since a drilled question about password expiry forces the assistant to commit to one of the two conflicting passages.

5 Discussion

Three observations follow from the case study, all of them descriptive.

Retrieval duality is a quality problem, not only a convenience problem. The two forms of the retrieved title are not redundant. A 12.2 MB PDF is a poor revision surface; a note tree is a poor provenance surface. Any vault-scale study system must decide, per artifact, which form is authoritative, and this decision is not recorded anywhere in the vault itself. The case

study’s resolution—note tree as study surface, PDF as provenance reference—is a reasonable default but is a convention adopted here, not a property discovered in the material.

Objective alignment is auditable at the note level. Because the module states its three objectives explicitly and because its content decomposes cleanly onto them, a completeness check is possible: every content block in Table 3 maps to O1, O2, or O3, and the passkey and biometric blocks together show that O1 is the load-bearing objective. A note that omits its stated objectives cannot be audited this way, which is an argument for requiring objectives in extracted notes generally.

Internal contradiction is the dominant defect class. Only one of the two defects found was structural, and the structural one is self-announcing. The substantive defect was a semantic contradiction between two normative statements, which is precisely the class that survives into a learner’s belief system undetected. A keyword-level or rendering-level extraction check would catch the malformed question; nothing short of a semantic pass catches the expiry contradiction. This is the strongest argument in the case study for spending audit effort on reasoning coherence rather than on formatting.

Limitations. Four limitations bound these findings. The study covers one artifact from one vault on one machine. The module was audited descriptively; no learner was tested against it, so nothing here speaks to whether the module teaches effectively. The source PDF was not read, so the audit cannot distinguish note-internal errors from extraction-introduced ones. And the taxonomy in Table 3 records coverage, not correctness: a block being present in the note is not evidence that the note states it correctly.

6 Conclusion

A case-insensitive recursive search of a personal Obsidian vault resolved one CompTIA study title into two artifacts, a 12.2 MB source PDF and a fifteen-module extracted note tree, and the note tree was selected as the study surface. Module 7, “Identity and Access Management,” aligned to SY0-701, was read in full at 471 lines and 21.6 KB and found to cover authentication credential types, authentication best practices, and access control schemes across thirteen content blocks spanning the seven elements of authenticity, password architecture and an eight-class attack taxonomy, multifactor token types, hardware tokens, biometric modalities including keystroke dynamics, digest protection with bcrypt, PBKDF2, and Argon2, the single sign-on technology set, passkeys, and five rank-ordered access control schemes. The quality audit found two defects: an unreconciled contradiction on password expiry and a review question with two textually identical options. The session concluded with a drill protocol in which learner-supplied examination questions are answered with a correctness score and a line-level citation into the source note. Extending the audit by reading the source PDF against the note, and by line-resolving both defect sites, are the immediate next steps.

References

- [1] **CompTIA Security+ Guide to Network Security Fundamentals**, Mark Ciampa. Source PDF held in the personal Obsidian vault at `~/Documents/vmain`, size 12.2 MB. Author and title as recorded in the vault artifact.

- [2] **Module 7: Identity and Access Management**. Extracted study note, 471 lines, 21.6 KB, in a directory of fifteen extracted per-module notes under `~/Documents/vmain`. Aligned to the CompTIA SY0-701 objective domain; states three learning objectives.